

▲ LEG 4 · AUTOMOTIVE AI · DAYS 24–30

THE AI ROAD TRIP LEG 4

7 new self-driving AI papers.
Rebuilt from scratch in 7 days.

D24

D25

D26

D27

D28

D29

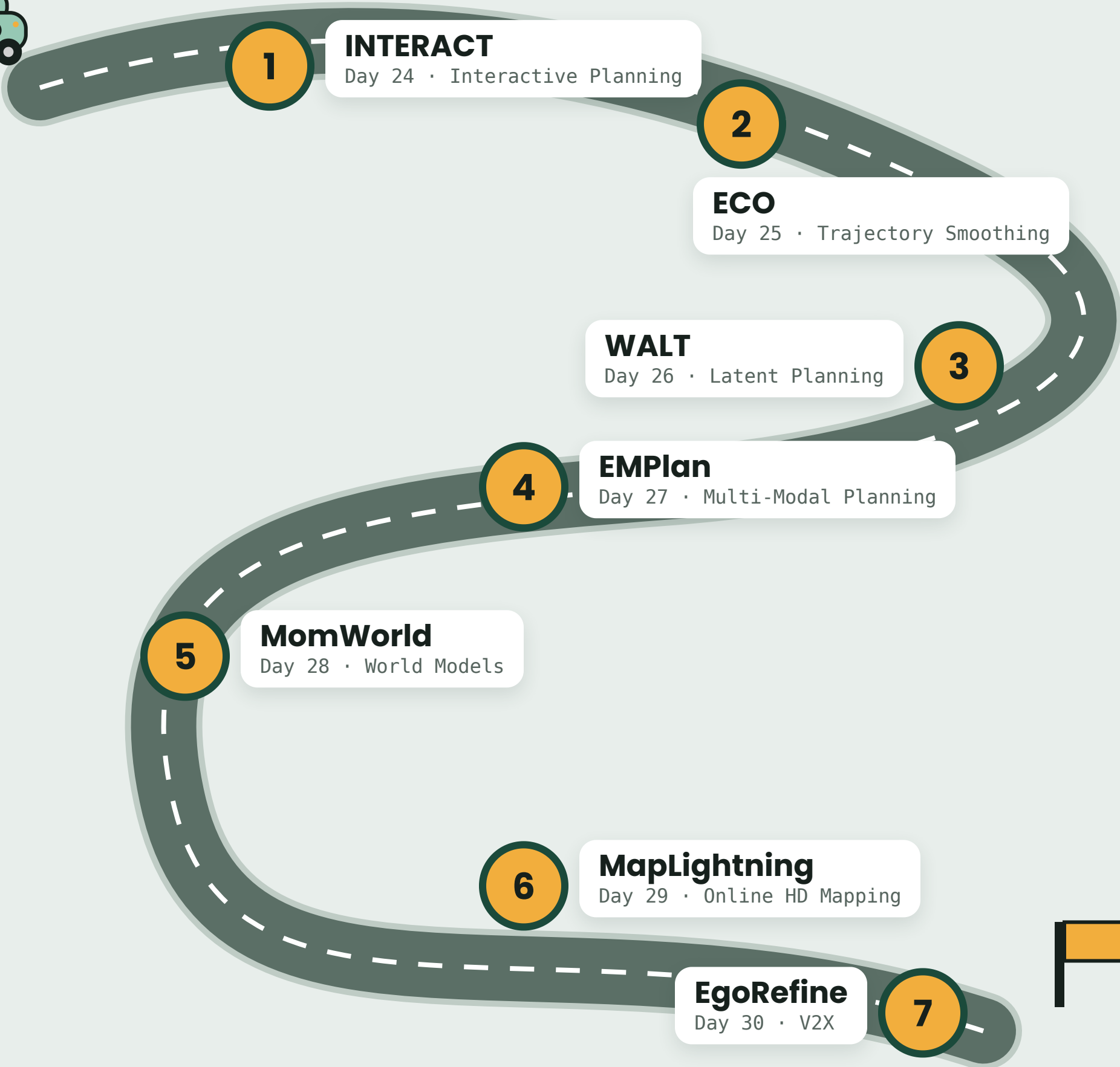
D30



Swipe to take Exit 01 →

This week's route

Planning · World models · HD maps · V2X. Seven exits, one repo.



7

PAPERS

7

PYTORCH REBUILDS

7

SIMULATION GIFS

30

SYSTEMS SO FAR

Swipe for Exit 01 →

01

EXIT 01 · INTERACTIVE PLANNING

INTERACT · arXiv 2609.31137

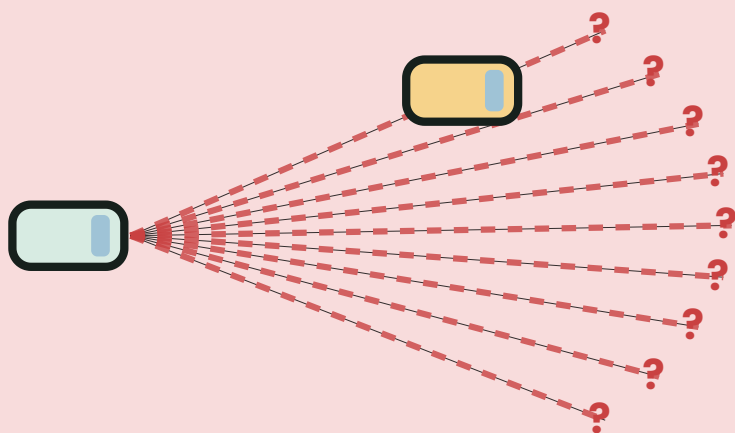
A. Distelzweig, A. Look, F. Janjoš ... J. Boedecker · Bosch · Univ. Freiburg · Coburg Univ.

Asks how others will react, once per idea, not once per wiggle

To plan a merge, a car must predict how its neighbours will respond. This AI predicts once per high-level intent (merge, keep, yield), then fine-tunes the path only where that forecast still holds.

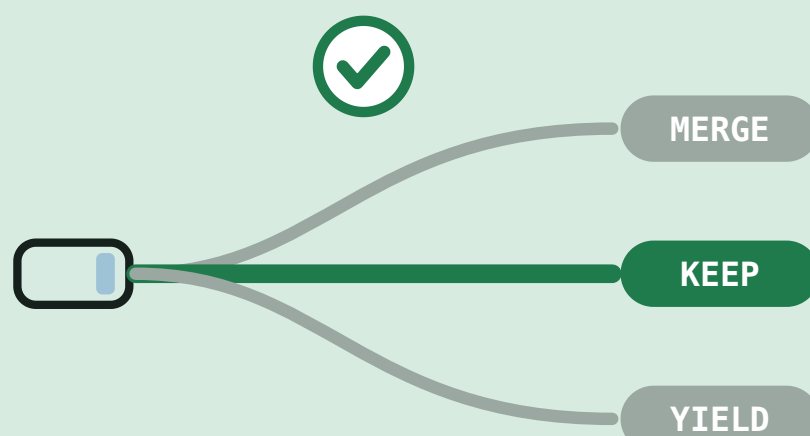
Like deciding “merge or yield” first, then steering, instead of re-asking the other driver for every tiny nudge.

OLD



ask the other car 640×

NEW



ask once per intent: 5×

128×

fewer prediction calls in my rebuild (5 vs 640)

✓ REPRODUCED IN MY REBUILD

02

EXIT 02 · TRAJECTORY SMOOTHING

ECO · arXiv 2609.31383

B. Zhang, M. Golchoubian ... B. Ivanovic, K. Chitta · U. Toronto · Vector · NVIDIA

Irons out a jittery plan without moving the destination

Driving AIs often output wobbly waypoints a real controller can't follow. This training-free add-on keeps the predicted endpoint, anchors to where the car has actually been, and redraws the points in between.

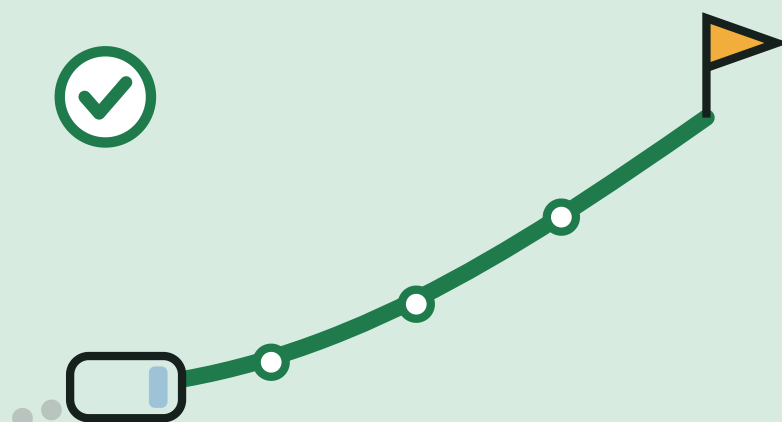
Like a GPS that keeps your destination but redraws a squiggly route as one smooth line.

OLD



jittery waypoints

NEW



same endpoint, smooth path

+71%

driving score on HUGSIM (paper) · -68% jerk in my rebuild

🕒 PARTLY REPRODUCED

03

EXIT 03 · LATENT PLANNING

WALT · arXiv 2609.30436

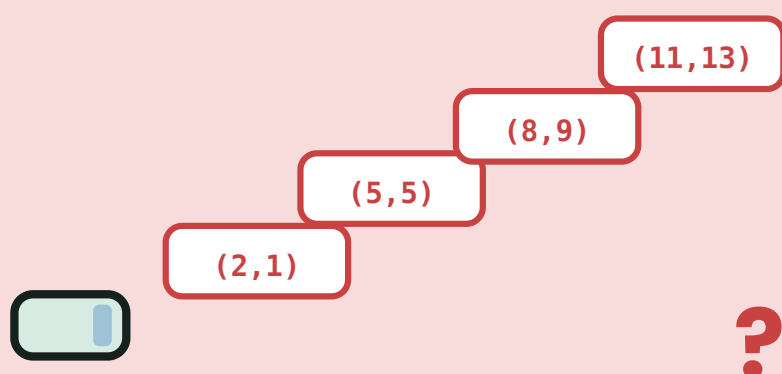
M. Jia, J. Guo ... P. Tan, W. Yin · HKUST · Horizon Robotics · CUHK

Plans in a world model's language, not raw coordinates

Instead of guessing raw x-y points, the planner writes its plan as a compact code borrowed from a pretrained world model that already understands how scenes move.

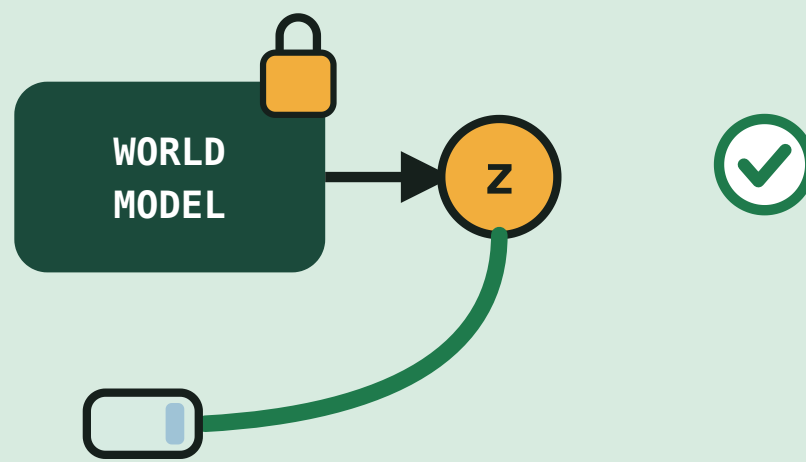
Like a navigator who thinks “take the next exit”, not in GPS coordinates.

OLD



raw x,y guesses

NEW



plan in a learned code z

-30.5% planner compute on NAVSIM (paper)

× NOT AT TOY SCALE

04

EXIT 04 · MULTI-MODAL PLANNING

EMPlan · arXiv 2609.38862

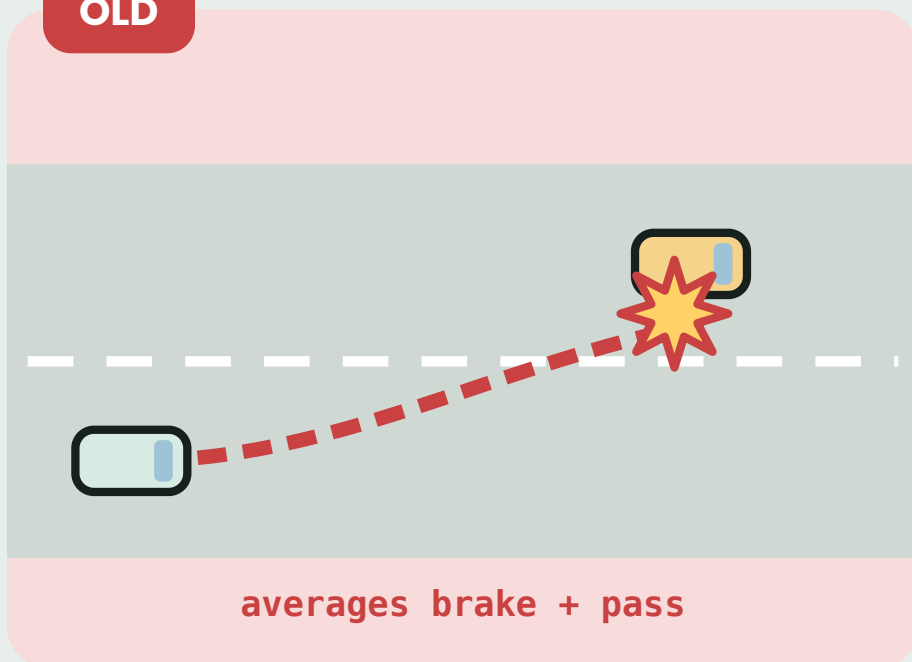
C. Chen, L. Wang, X. Zheng, J. Ma, H. Li

Picks brake OR overtake, never an unsafe average of both

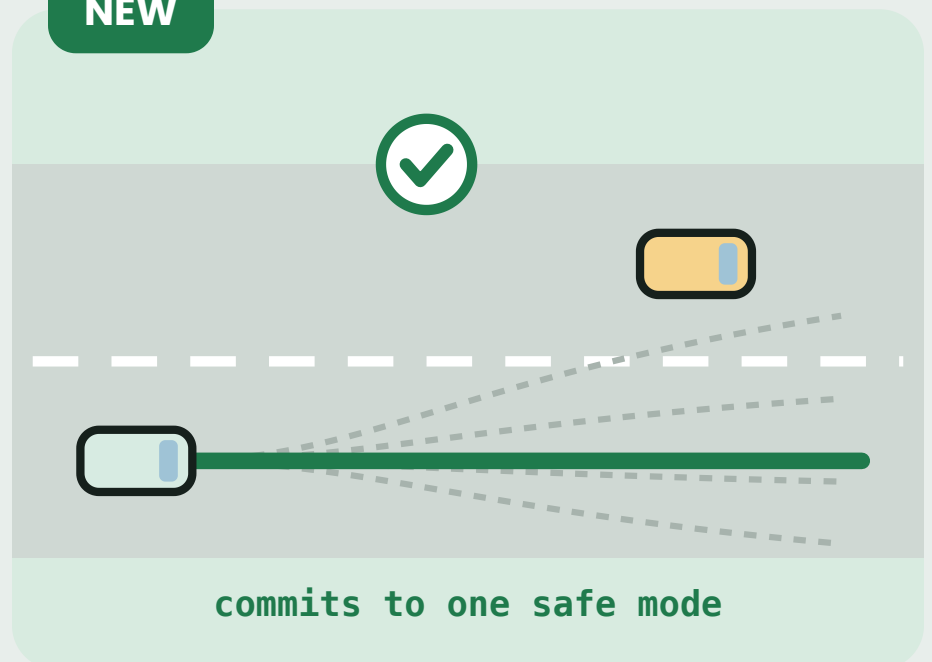
When experts sometimes brake and sometimes overtake, a single-path model averages them, straight into the car ahead. This AI keeps several candidate paths, refines each, then tunes them with simple safety rules.

Like choosing a lane decisively instead of straddling the line.

OLD



NEW

**53.9 → 2.5%**

collision rate in my rebuild (3 seeds)

✓ REPRODUCED IN MY REBUILD

05

EXIT 05 · WORLD MODELS

MomWorld · arXiv 2609.33737

Z. Song, S. Zhang ... C. Lv · NTU · BJTU · DUT · Tsinghua

Remembers where traffic was heading, and forgets on cue

Six-second forecasts tend to lose the motion trend. This world model carries a “momentum” memory forward, plus a reset gate that drops it the moment the scene changes, like a lead car braking.

Like riding momentum downhill, but braking the instant the light turns red.

OLD

old trend: go!



stale momentum

NEW



RESET



gate drops outdated momentum

-12.2% collision rate vs. MomAD, 6 s horizon (paper)

× NOT AT TOY SCALE

06

EXIT 06 · ONLINE HD MAPPING

MapLightning · arXiv 2610.01905

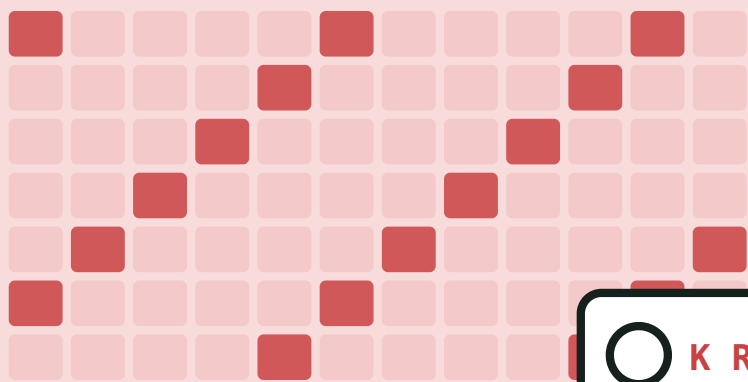
S. Zheng, A. Ghosh, M. Ramanagopal, S. Narasimhan · CMU

Sketches the lane map from just 24 tokens

Building a lane-level map on the fly usually needs a dense bird's-eye grid and exact camera calibration. This AI squeezes the whole map into 24 learned tokens, with no camera parameters at all.

Like sketching a road in a few confident strokes instead of painting every pixel.

OLD

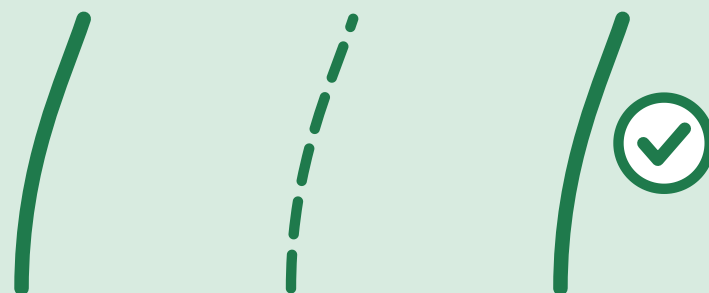


dense grid + exact calibration

NEW



24 map tokens



vector lanes, no camera params

16.7× fewer tokens, +10.1 mAP on nuScenes (paper)

🔗 PARTLY REPRODUCED

07

EXIT 07 · V2X · COOPERATIVE PERCEPTION

EgoRefine · arXiv 2610.00319

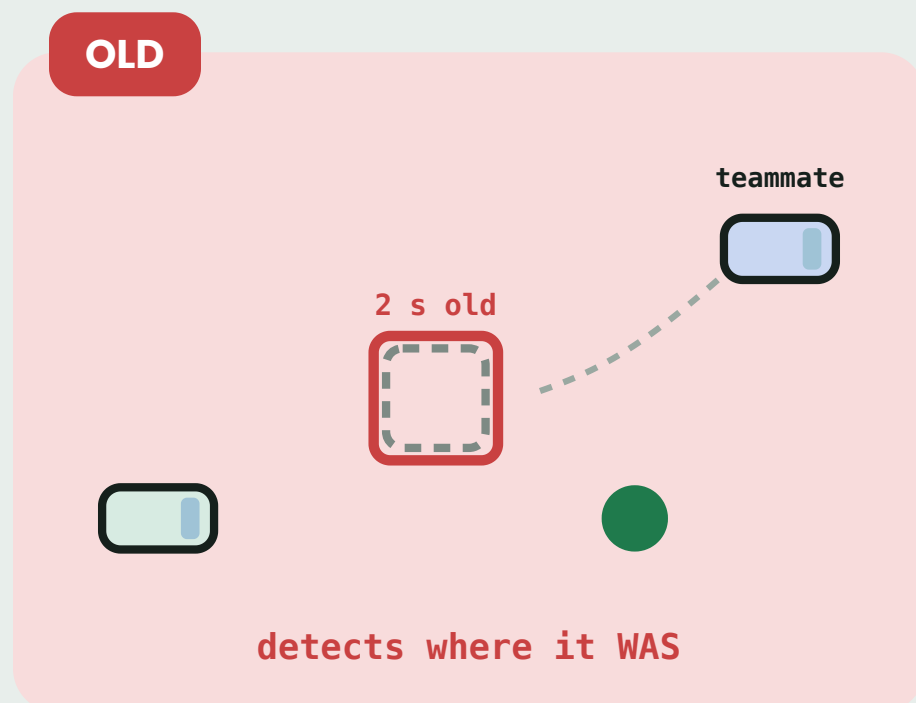
L. Kong, Y. Zang ... K. Yang, Z. Li · Hunan University

Fixes a teammate's late message with what it sees now

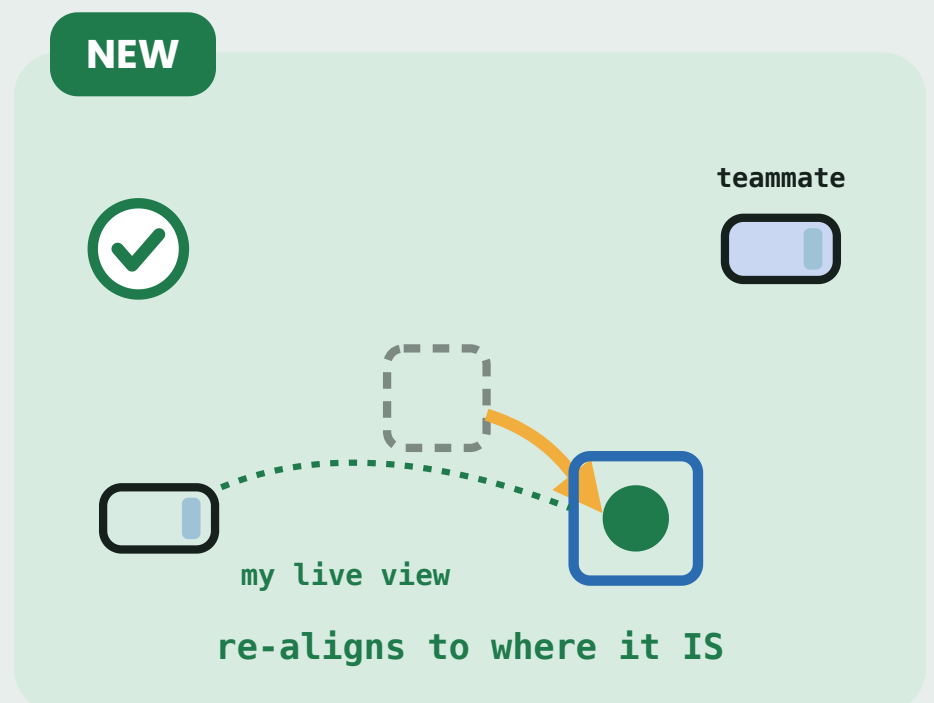
Connected cars share what they see, but messages arrive late and objects have moved on. This AI predicts where to look in the old message, and uses its own live view to correct that guess.

Like hearing “the cyclist was at the corner two seconds ago”, then checking your mirror to update.

OLD



NEW

**+1.6 AP**

vs. TraF-Align on V2V4Real + DAIR-V2X (paper)

🕒 PARTLY REPRODUCED

REALITY CHECK

The honest scorecard

I rebuild every paper from scratch in PyTorch on synthetic data. Here's what held up, and what didn't.

2

HELD UP

3

PARTLY

2

NOT AT TOY SCALE

01

INTERACT

Paper: SOTA on nuPlan & interPlan

My rebuild: 128× fewer predictor calls; -82.6% forecast error

✓

Held up

02

ECO

Paper: +71% HD-Score on HUGSIM

My rebuild: -68% jerk, but lateral error 0.35 → 0.45 m

●

Partly

03

WALT

Paper: -30.5% compute, +0.4 PDMS

My rebuild: latent ADE 3.48 m vs 3.25 m raw baseline

×

Not at toy scale

04

EMPlan

Paper: Real-time multi-modal planning

My rebuild: collisions 53.9% → 2.5% (3 seeds)

✓

Held up

05

MomWorld

Paper: -12.2% collisions vs MomAD

My rebuild: ADE 0.46 m vs 0.42 m baseline

×

Not at toy scale

06

MapLightning

Paper: +10.1 mAP, 16.7× fewer tokens

My rebuild: robustness edge only at 6° jitter

●

Partly

07

EgoRefine

Paper: +1.6 AP@0.5 vs TraF-Align

My rebuild: collab +11 AP; alignment gain ≈ noise

●

Partly

A toy-scale miss isn't a verdict on the paper. It's a signal about what the idea needs: scale, real data, or details the abstract doesn't show.

30 SYSTEMS • 30 DAYS

Want the code?

Tested PyTorch implementations, simulation GIFs and honest write-ups for all 30 systems so far. Free and open on GitHub.

`github.com/Lalithprabu/
automotive-ai-daily-reviews`

★ **Star the repo**

Follow for Leg 5 →

